# Effect size and practical importance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

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WORKBOOK · 1 OF 15

## Recall and choose your starting point

> Record your starting explanation for Effect size and practical importance before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Uncertainty intervals

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Confounders and controls

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Hypotheses and measurable outcomes

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## Practise with fading help

> Complete the supported Effect size and practical importance trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Effect size** and practical importance. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

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## Individual reasoning check

> Check the proposed Effect size and practical importance explanation against the supplied facts before discussing it.

Use the worked case for **Effect size** and practical importance. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 15

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Apply a new case independently

Baseline6, 8, 10; candidate5, 7, 9. Useful mean gain at least1.5. Decide.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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## W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Effect size and practical importance facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

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## W2 Trace the worked case

> Trace the worked case. Apply the supplied Effect size and practical importance facts and keep your reasoning.

Baseline times10, 12, 8; candidate8, 9, 7 seconds. Useful gain threshold2 seconds. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

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## W3 Audit the limit

> Audit the limit. Apply the supplied Effect size and practical importance facts and keep your reasoning.

A report omits this qualification: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

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## W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Effect size and practical importance facts and keep your reasoning.

Baseline6, 8, 10; candidate5, 7, 9. Useful mean gain at least1.5. Decide.

**My reasoning and evidence**

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## W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Effect size and practical importance facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

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## Project checkpoint and reflection

> Connect Effect size and practical importance to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled **effect size** and practical importance **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **effect size** and practical importance.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

**My project change, supporting evidence and remaining limit**

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WORKBOOK · 11 OF 15

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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## Pictorial case practice: Effect size and practical importance

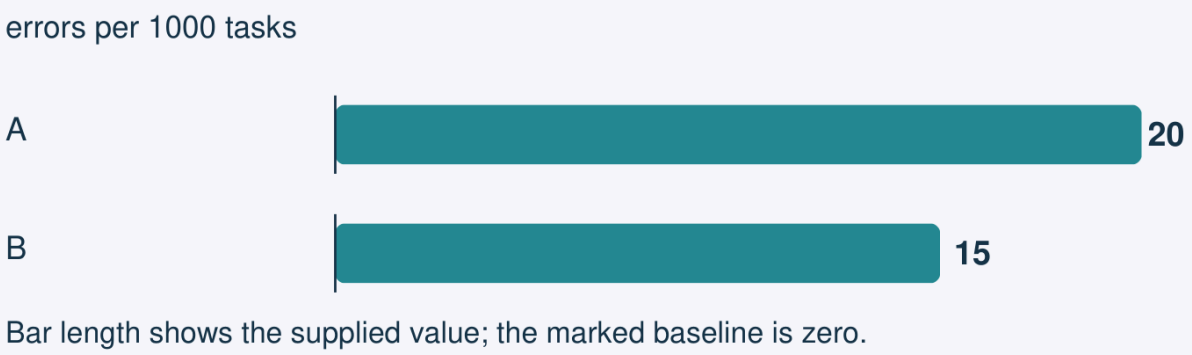
> An improvement with a practical threshold. A team compares error rates against a predeclared practical threshold. Explain the labelled step before checking the changed condition.

An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.

A makes 20 errors in 1000 tasks; B makes 15. Practical improvement requires at least 10 fewer errors per 1000.

AI ACADEMY / AI-7/13

## Errors on 1000 tasks




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/13

# Effect size and practical importance

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Compute rates

A is 2%; B is 1.5%.



### 2. Compute effect size

B has 5 fewer errors per 1000, a 0.5-percentage-point reduction.



### 3. Apply practical threshold

Five is less than ten, so the observed change misses this threshold.

**CHANGE ONE THING**

B makes 8 errors in 1000.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Five is less than ten, so the observed change misses this threshold.

1. Apply **practical threshold**

2. Compute rates

3. Compute **effect size**

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply **practical threshold** Five is less than ten, so the observed change misses this threshold. **Effect size** describes how much the outcome changes.

Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: B makes 8 errors in 1000. Which conclusion is supported by the supplied facts?

1. The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: The reduction becomes 12 per 1000 and meets the stated **practical threshold**. Twenty minus eight exceeds the required ten fewer errors.

### Repair a missing building block

Review Uncertainty intervals (ai-7/12). Goal: Explain the **mechanism** and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2].

Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Review Confounders and controls (ai-7/10). Goal: Explain the **mechanism** and terminology of confounders and controls. Short practice: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the **mechanism** and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

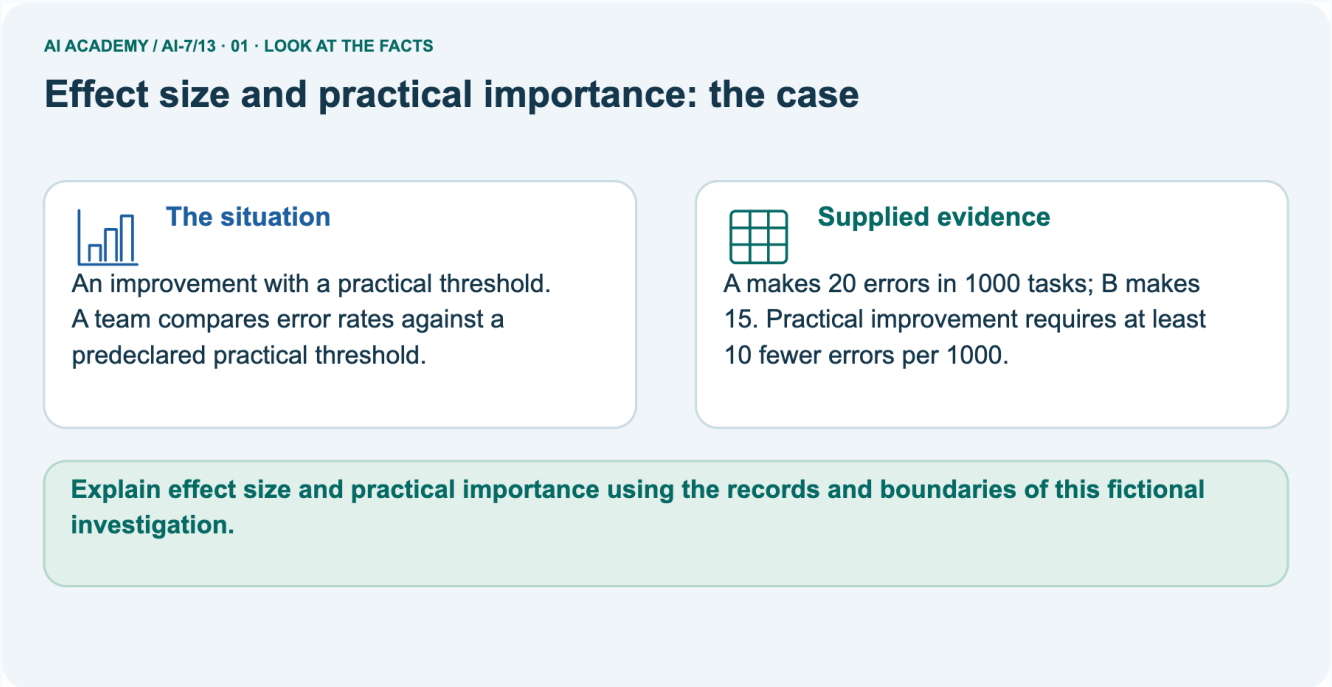
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WORKBOOK · 15 OF 15

# Visual concept journey

> Explain each Effect size and practical importance picture from its labels, then record a relevant limit.

### 01. Effect size and practical importance: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

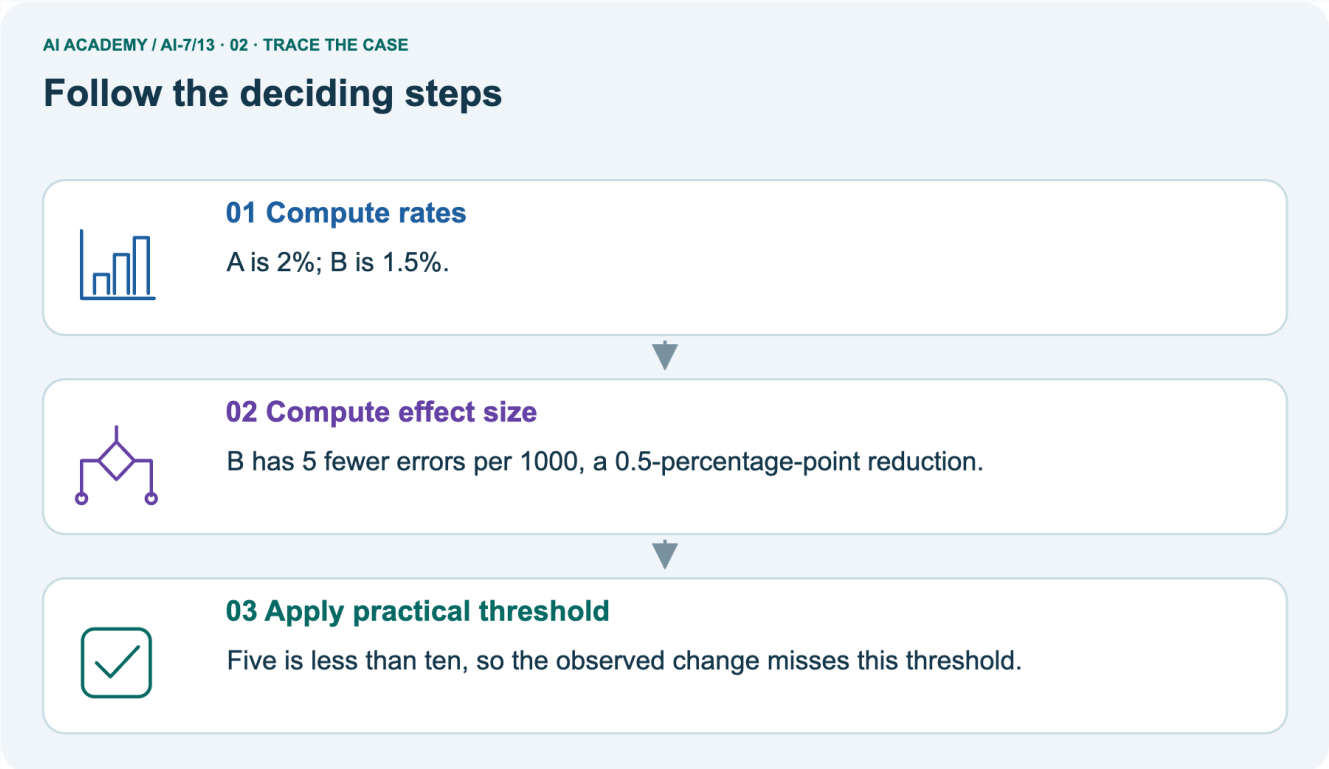
An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**. Supplied facts: A makes 20 errors in 1000 tasks; B makes 15.

Practical improvement requires at least 10 fewer errors per 1000.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Compute rates: A is 2%; B is 1.5%. → Compute **effect size**: B has 5 fewer errors per 1000, a 0.5-percentage-point reduction. → Apply **practical threshold**: Five is less than ten, so the observed change misses this threshold.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

### 03. Errors on 1000 tasks



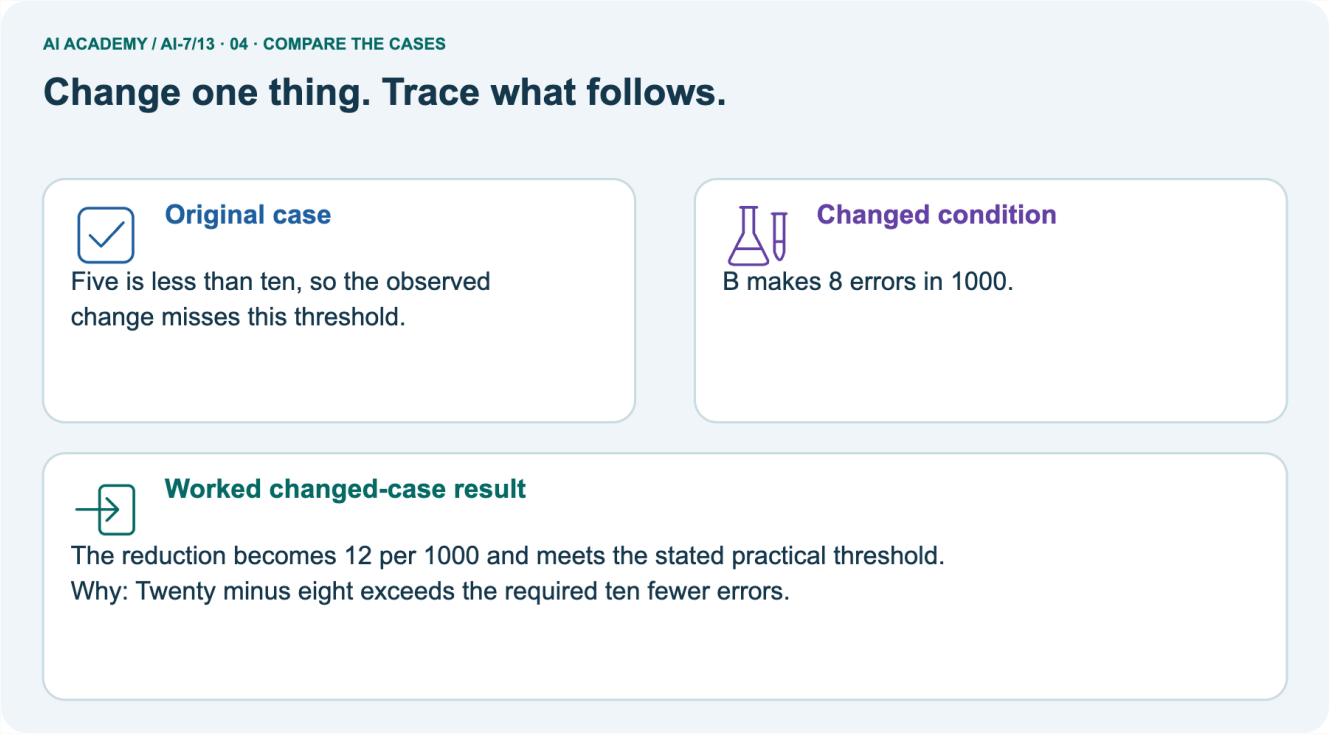

Follow the labelled picture with the matching explanation. Open the vector diagram

Errors on 1000 tasks. Units: errors per 1000 tasks. A: 20. B: 15.

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

### 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

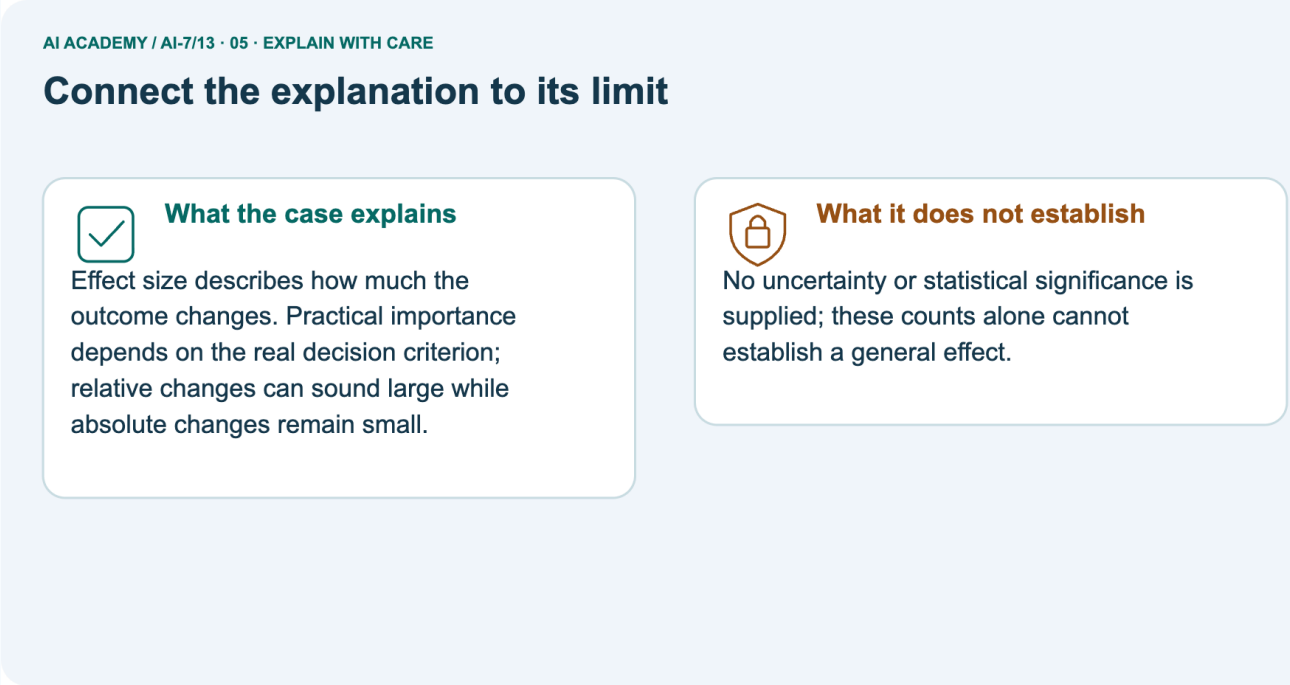
Original case: Five is less than ten, so the observed change misses this threshold. Changed condition: B makes 8 errors in 1000. Expected result: The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

Reason: Twenty minus eight exceeds the required ten fewer errors.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

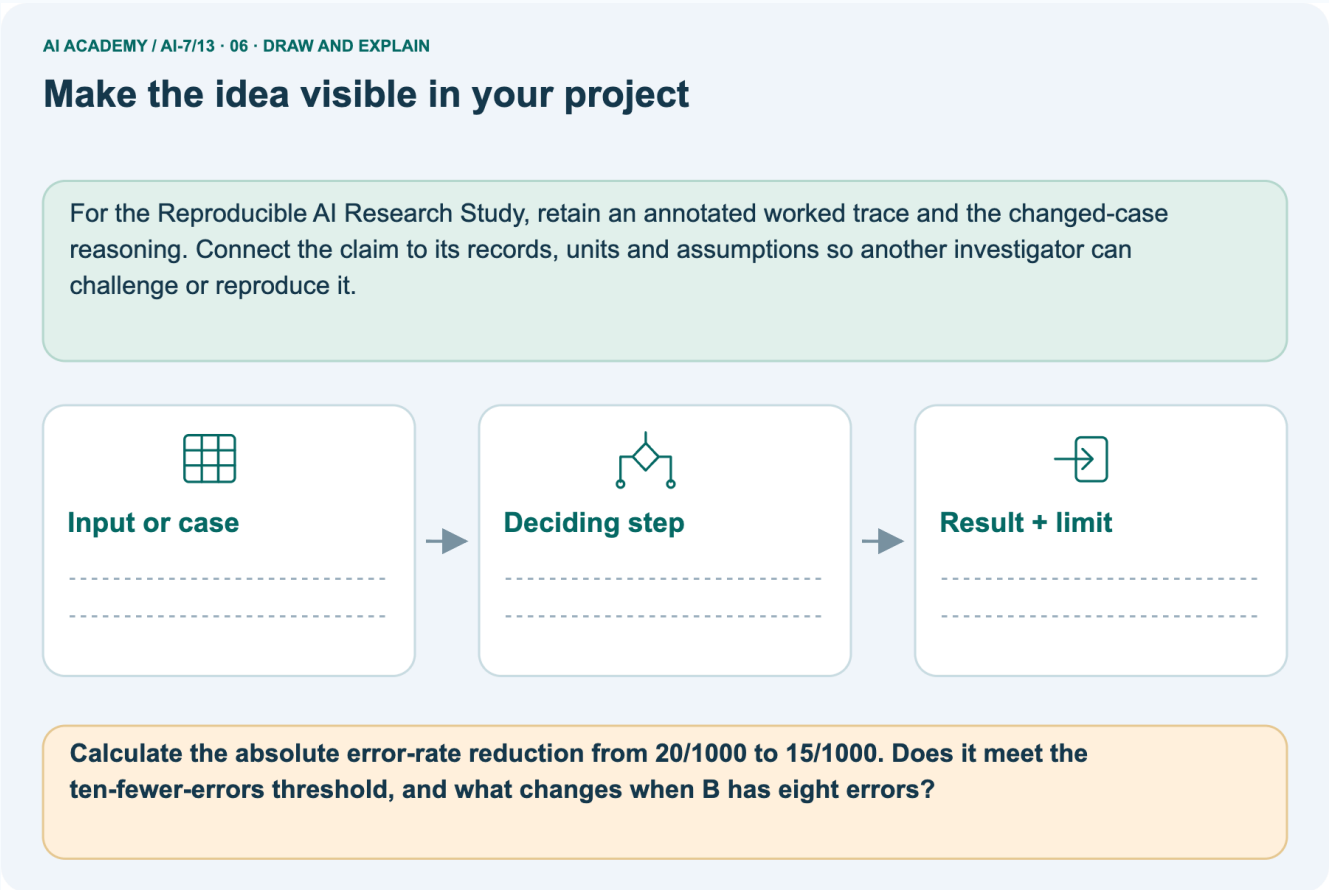
**Effect size** describes how much the outcome changes. Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

Limit: No uncertainty or statistical significance is supplied; these counts alone cannot establish a general effect.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

Draw your project trace and justify the deciding step using **evidence**.

**My picture explanation, deciding evidence and one limit**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.